

K-FIELD

Carbon-Fiber Tube Laboratory Experiment

Identical 10 mm by 8 mm by 100 mm tubular geometry at a 1600 ft/s outer-rim velocity: mass, spin state, K-Field force, fixed-scale projection, normalized force, and spin-reversal envelope

K-Field carbon-fiber tube fixed-laboratory experiment

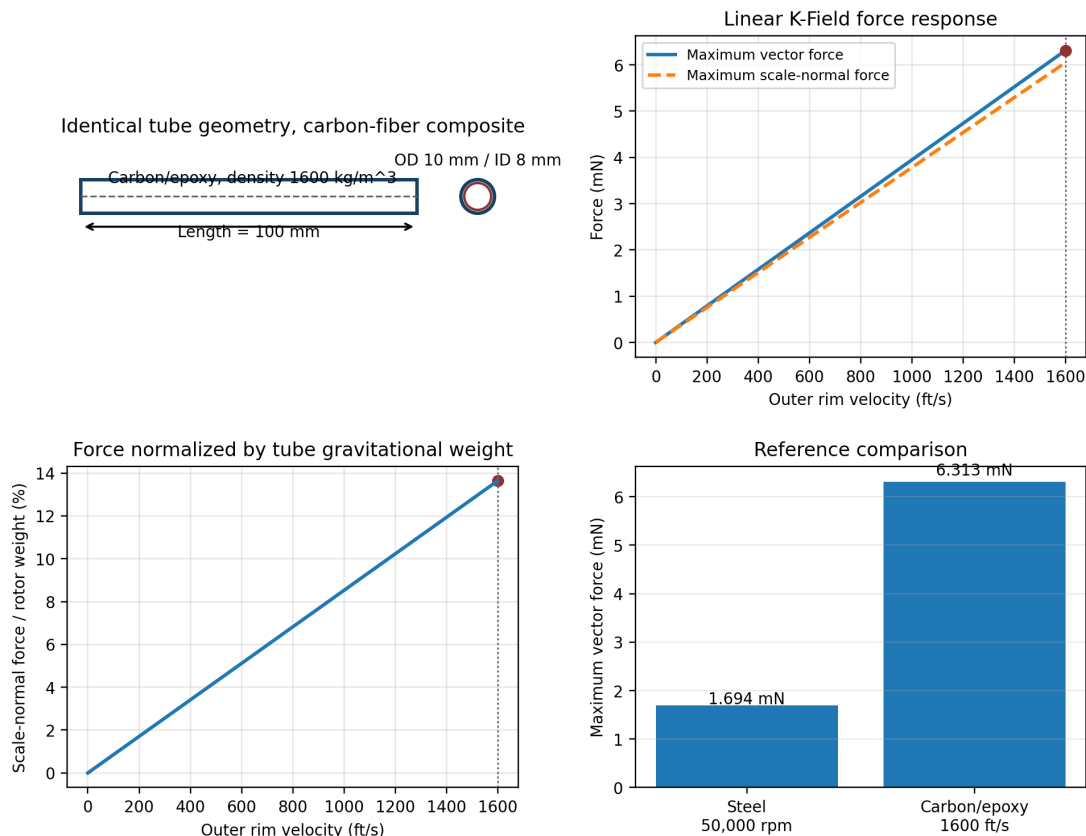


Figure 1. Title composite. Every plotted quantity is generated from the exact tube geometry, the 1600 kg/m^3 carbon/epoxy density record, the supplied Cell3 response ledger, and the stated 1600 ft/s outer-rim velocity.

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 Archival computational edition - Unix epoch 1784915745

Frontispiece: authoritative Cell3 base cell, face planes, and all nodes

Authoritative Cell3 base cell: six faces, 12 edges, all 27 nodes

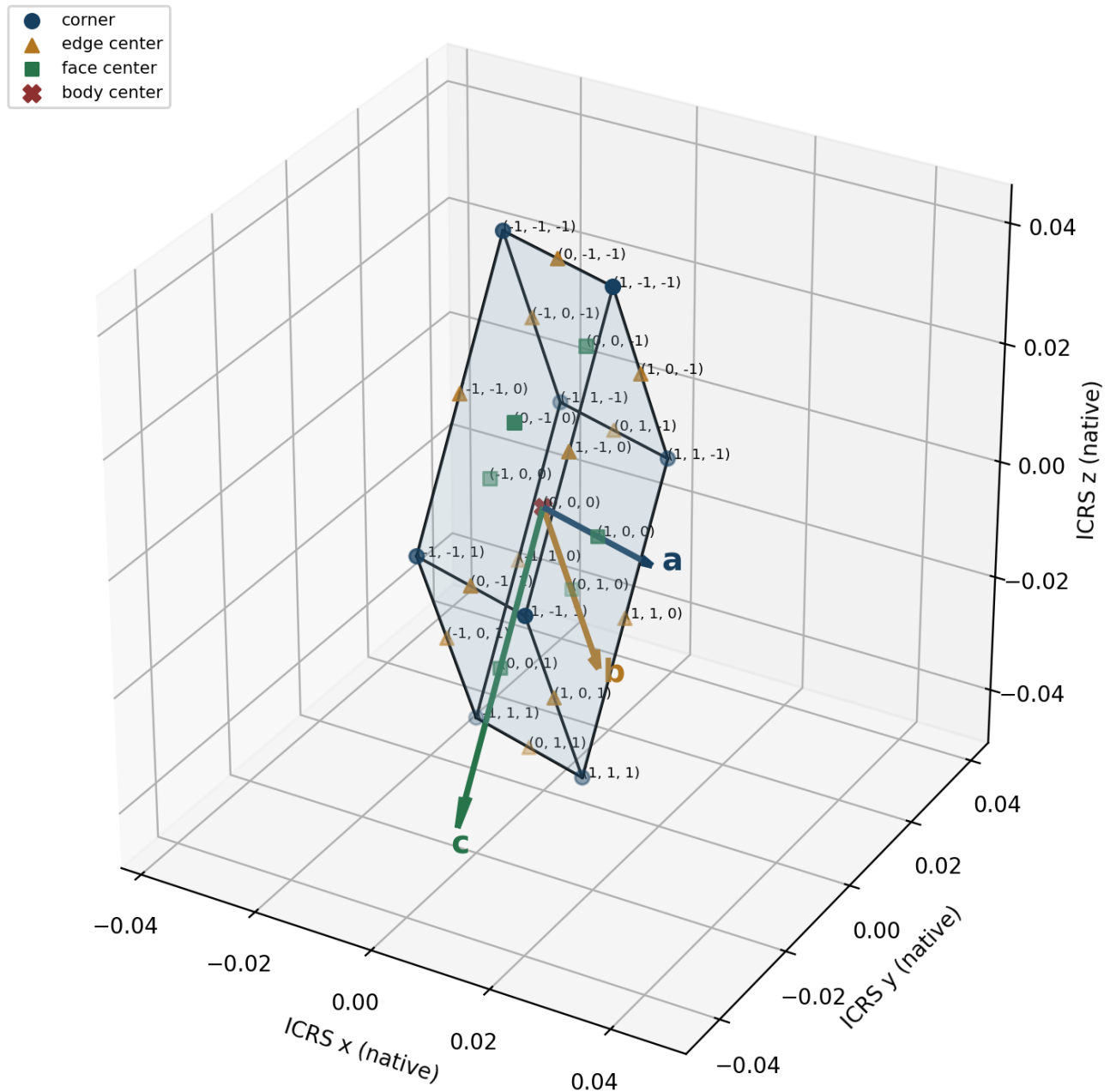


Figure 2. Authoritative Cell3 base cell in ICRS/J2000 Cartesian coordinates. The centered primitive cell is reconstructed from the source basis matrix. All six boundary faces, 12 edges, 27 half-step address nodes, integer node labels, and basis vectors are rendered in native dimensions.

Cell3 face-plane system

Cell3 central face planes and unit normals (native ICRS dimensions)

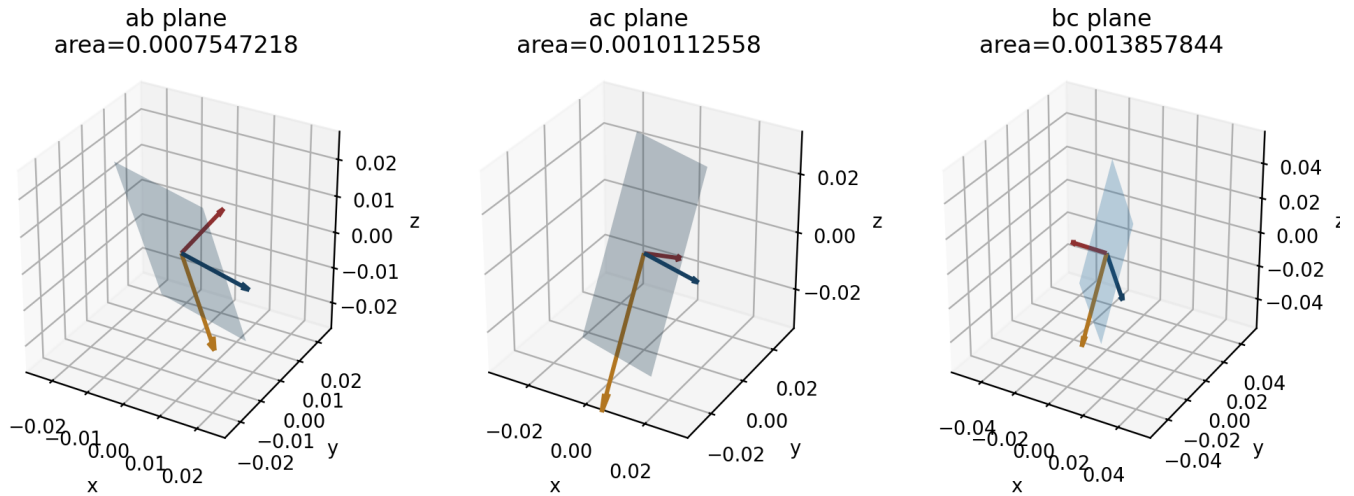


Figure 3. Central ab, ac, and bc planes generated from the exact basis vectors. Each panel reports the source face area and uses the source unit normal. Coordinates are native ICRS lengths.

The Cell3 geometry fixes the directional response used in the force calculation. It is therefore part of the complete experimental definition rather than a decorative representation.

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This paper is a standalone educational and archival reference. It derives the carbon-fiber tube mass from commercially documented density, converts the required 1600 ft/s outer-rim velocity into angular speed and revolutions per minute, computes inertia, angular momentum, stored energy, centripetal acceleration, maximum K-Field force, maximum fixed-scale force, apparent mass modulation, spin-reversal separation, and force normalized by rotor gravitational weight. It also preserves the complete Cell3 geometry, source field matrix, source vectors, all 27 nodes, reconstruction equations, and numerical ledgers needed to reproduce the result.

Abstract

An identical tubular geometry with outside diameter 10 mm, inside diameter 8 mm, wall thickness 1 mm, and length 100 mm is evaluated using a carbon/epoxy density of 1600 kg/m³. The resulting tube mass is 4.523893 g. An outer-rim velocity of 1600 ft/s equals 487.680 m/s and requires angular speed 97536.000 rad/s, or 931,400.2 rpm, at the 5 mm outer radius. Applying the integrated tube law $F_K = (\kappa_F m \omega / 2)(n \times g)$ gives a maximum complete force of 6.312865 mN and a maximum fixed-laboratory scale-normal force of 6.048933 mN. The one-direction apparent-weight amplitude is 0.616820 g, while direct reversal between equal positive and negative axial spin produces a 1.233639 g scale separation. The maximum scale-normal force is 13.634705% of the tube gravitational weight, and the complete reversal separation is 27.269411% of tube weight.

Central result

At the optimum fixed-laboratory projection, the carbon/epoxy tube produces a scale-normal force of 6.048933 mN, equivalent to 0.616820 g in either direction. Reversing axial spin immediately moves the scale between the two opposite states, producing the full 12.097867 mN or 1.233639 g separation without waiting approximately 12 sidereal hours for Earth rotation.

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1. Scope, definitions, and source lock

The mechanical object is one complete hollow circular tube rotating about its longitudinal axis. The tube dimensions are identical to the supplied surgical-stainless reference: 10 mm outside diameter, 8 mm inside diameter, 1 mm radial wall thickness, and 100 mm length. Only the material density and operating speed are changed. The material is carbon-fiber reinforced epoxy, represented by the commercially documented density 1.60 g/cc. The outer cylindrical surface is assigned a tangential velocity of 1600 ft/s.

All field quantities are retained from the supplied K-Field tubular-rotor source ledger. The ICRS/J2000 coordinate system, Cell3 basis, G_phase matrix, Earth velocity vector, effective response vector $\mathbf{g}$, and coupling constant κ_F remain unchanged. This isolates the effects of tube density and axial angular speed.

$$\begin{aligned} \mathbf{g} &= \mathbf{G_phase} \mathbf{V_E} \\ \mathbf{F_K} &= (\kappa_F m \omega / 2) (\mathbf{n} \times \mathbf{g}) \\ \mathbf{a_K} &= \mathbf{F_K} / m = (\kappa_F \omega / 2) (\mathbf{n} \times \mathbf{g}) \end{aligned}$$

The signed angular speed ω carries the direction of axial rotation. Replacing ω by $-\omega$ reverses the complete force vector while preserving its magnitude.

Quantity	Value	Role
Earth velocity $\mathbf{V_E}$	(np.float64(-359003.199065813), np.float64(76679.886034002), np.float64(-44701.060100688))	m/s, ICRS
Field response $\mathbf{g}$	(np.float64(-880221.981635091), np.float64(-143911.436561599), np.float64(266335.026948619))	effective response vector
$ \mathbf{g} $	930825.217270434834	effective response magnitude
Equatorial $ \mathbf{g_xy} $	891908.761324178427	fixed-scale maximum projection
Projection factor	0.958191446445	$\mathbf{g_xy} / \mathbf{g} $
κ_F	3.07405089970203453e-11	recommended coupling
κ_F interval	[3.04273994475464998e-11, 3.10536185464941972e-11]	calibration interval

2. Commercial carbon/epoxy density basis

The computation uses the physical-property record titled Goodfellow Carbon/Epoxy Composite Tube. The listed density is 1.60 g/cc, equivalent to 1600 kg/m³ and 0.0578 lb/in³. The value is a finished carbon-fiber/epoxy composite tube density rather than the density of bare carbon fiber alone.

$\rho_{CF} = 1.60 \text{ g/cm}^3$
 $\rho_{CF} = 1600 \text{ kg/m}^3$

Field	Record
Material	Carbon fiber - epoxy resin matrix composite tube
Commercial record	Goodfellow Carbon/Epoxy Composite Tube
Density	1.60 g/cc = 1600 kg/m ³
Source database	MatWeb
Reference	https://www.matweb.com/search/DataSheet.aspx?MatGUID=51313a82bc67483abf4f52c0e2934c88
Use in this report	Mass, inertia, angular momentum, energy, and absolute force

The free acceleration and the force-to-weight percentage do not depend on density under this force law. Density controls the tube mass and therefore controls absolute force, stored rotational energy, and angular momentum for fixed geometry and speed.

3. Identical tubular geometry and mass

Dimensionally exact modeled carbon/epoxy tube geometry

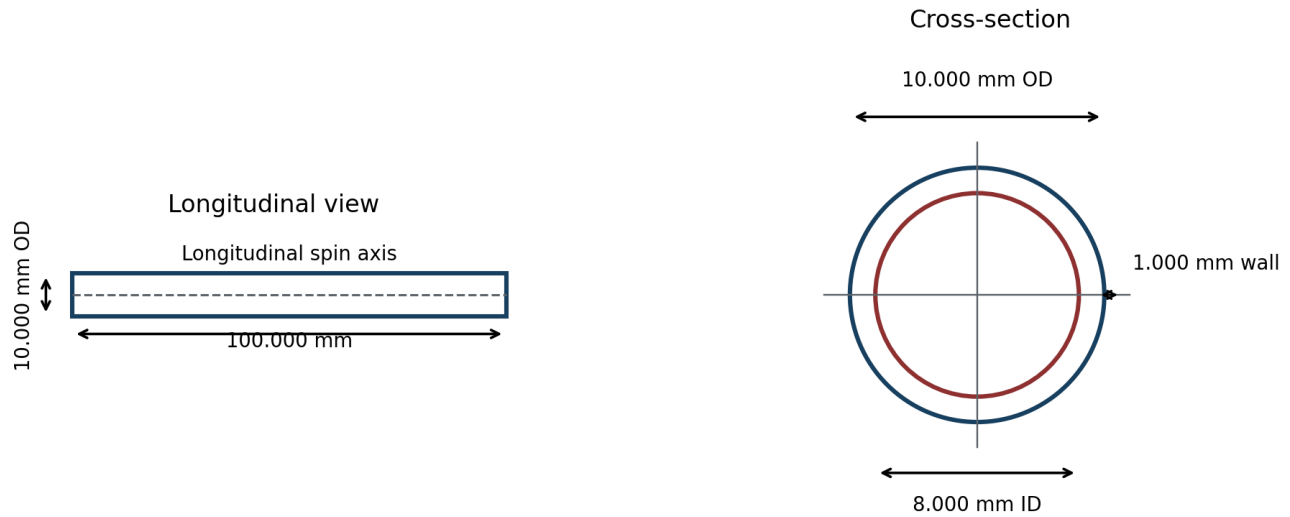


Figure 4. Dimensionally exact tube geometry used in the calculation. The hollow annular section is 10.000 mm OD, 8.000 mm ID, and 100.000 mm long.

The metal or composite material occupies the annular cross-section between the outer and inner radii. The volume and mass are

$$A = \pi (r_o^2 - r_i^2)$$

$$V = A L$$

$$m = \rho V$$

Quantity	Value
Outer radius r_o	0.005000000 m
Inner radius r_i	0.004000000 m
Wall thickness	0.001000000 m
Length L	0.100000000 m
Annular area	2.827433388230814e-05 m ²
Material volume	2.827433388230815e-06 m ³
Carbon/epoxy density	1600.000000 kg/m ³
Tube mass	0.004523893421169 kg = 4.523893421 g

The carbon/epoxy tube has exactly 20% of the source 316L tube mass because the selected density, 1600 kg/m³, is 20% of the source stainless density, 8000 kg/m³, while geometry is unchanged.

4. 1600 ft/s rim-speed conversion

Operating state for the 5 mm outer radius

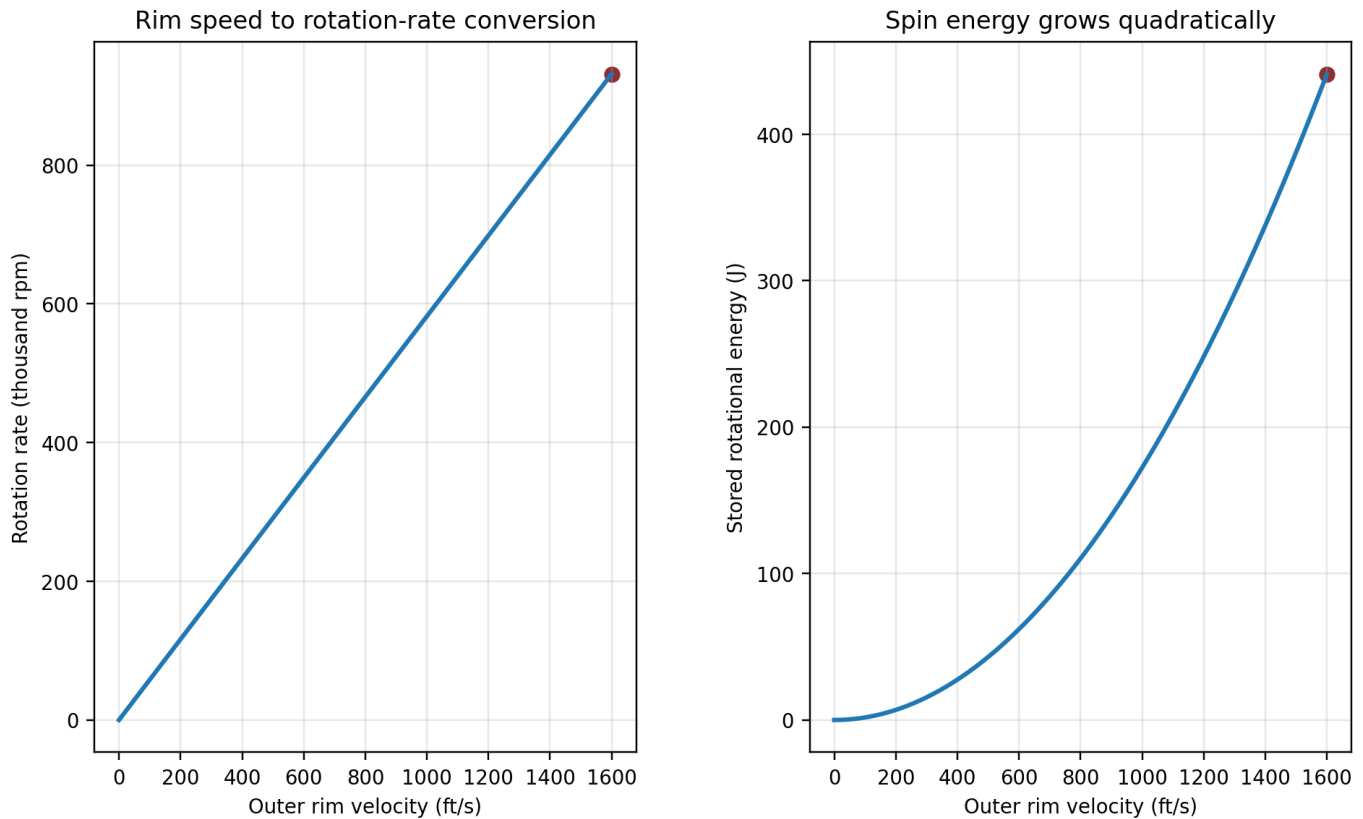


Figure 5. The left panel converts outer-rim speed to rotational rate for a 5 mm radius. The right panel shows the quadratic growth of stored rotational energy.

The requested outer-surface speed is converted using the exact foot-to-meter factor 0.3048 m/ft.

```
v_o = 1600 ft/s x 0.3048 m/ft = 487.68 m/s
omega = v_o / r_o
RPM = omega (60 / 2 pi)
```

Quantity	Result
Outer-rim velocity	1600.000 ft/s = 487.680000 m/s
Angular speed	97536.000000000 rad/s
Rotation rate	931400.191765 rpm
Rotation frequency	15523.336529411 Hz
Rotation period	6.441914069861e-05 s
Inner-surface velocity	390.144000 m/s

The required speed is 18.628003835 times the 50,000 rpm stainless reference angular speed.

5. Rotational inertia, momentum, and energy

For a uniform hollow circular tube rotating about its longitudinal axis, the axial mass moment of inertia is

$$I = (m/2) (r_o^2 + r_i^2)$$

$$H = I \omega$$

$$E_{\text{spin}} = (1/2) I \omega^2$$

Quantity	Result	Interpretation
Axial moment of inertia	9.273981513397073e-08 kg m ²	Resistance to angular acceleration
Specific axial inertia	2.050000000000000e-05 m ²	Geometry-only inertia per mass
Angular momentum	9.045470608906968e-03 kg m ² /s	Stored axial rotational momentum
Specific angular momentum	1.999488000000 m ² /s	Angular momentum per mass
Stored rotational energy	441.129510655175 J	Energy at 1600 ft/s outer rim
Specific rotational energy	97511.030784000002 J/kg	Energy per rotor mass

The K-Field acceleration depends directly on signed angular speed rather than independently on the inertia. Inertia remains essential to the practical drive system because it controls spin-up torque, angular momentum, and stored energy.

6. Centripetal kinematic state

Every material element of the tube wall follows a circular path around the longitudinal axis. The local centripetal acceleration is $a_c = \omega^2 r = v^2/r$. It varies linearly with radius for fixed angular speed.

Location	Radius	Tangential speed	Centripetal acceleration	Acceleration in standard g
Outer surface	0.005000 m	487.680000 m/s	4.756635648e+07 m/s ²	4.850418489e+06 g
Inner surface	0.004000 m	390.144000 m/s	3.805308518e+07 m/s ²	3.880334792e+06 g

These accelerations describe the mechanical rotor state and bearing/containment challenge. They are distinct from the predicted K-Field center-of-mass acceleration. A complete mechanical design requires laminate schedule, fiber orientation, manufacturing quality, balance, bearing dynamics, hub interfaces, containment, and material allowables that are not supplied by the density record. The present computation therefore defines the kinematics and K-Field force, not a structural certification for operation at 931,400 rpm.

7. Integrated K-Field force law

The complete circular wall contains all circumferential phases simultaneously. Integration around the tube removes a rotating pulse and leaves a steady center-of-mass force proportional to tube mass and signed axial angular speed.

$$\begin{aligned}F_K &= (\kappa_F m \omega / 2) (n \times g) \\|F_K| &= (\kappa_F m |\omega| / 2) |g| \sin(\beta) \\a_K &= F_K / m\end{aligned}$$

Here β is the angle between the rotor axis n and the field-response vector g . Maximum complete force occurs at $\beta=90$ degrees. Under this orientation the mass cancels from acceleration, but not from force.

Factor	Value
κ_F	3.07405089970203453e-11
Tube mass m	0.004523893421169 kg
Signed angular speed magnitude	97536.000000000 rad/s
$ g $	930825.217270434834
Maximum acceleration	1.395449549837 m/s ²
Maximum force	6.312865038083341e-03 N

8. Maximum complete force vector

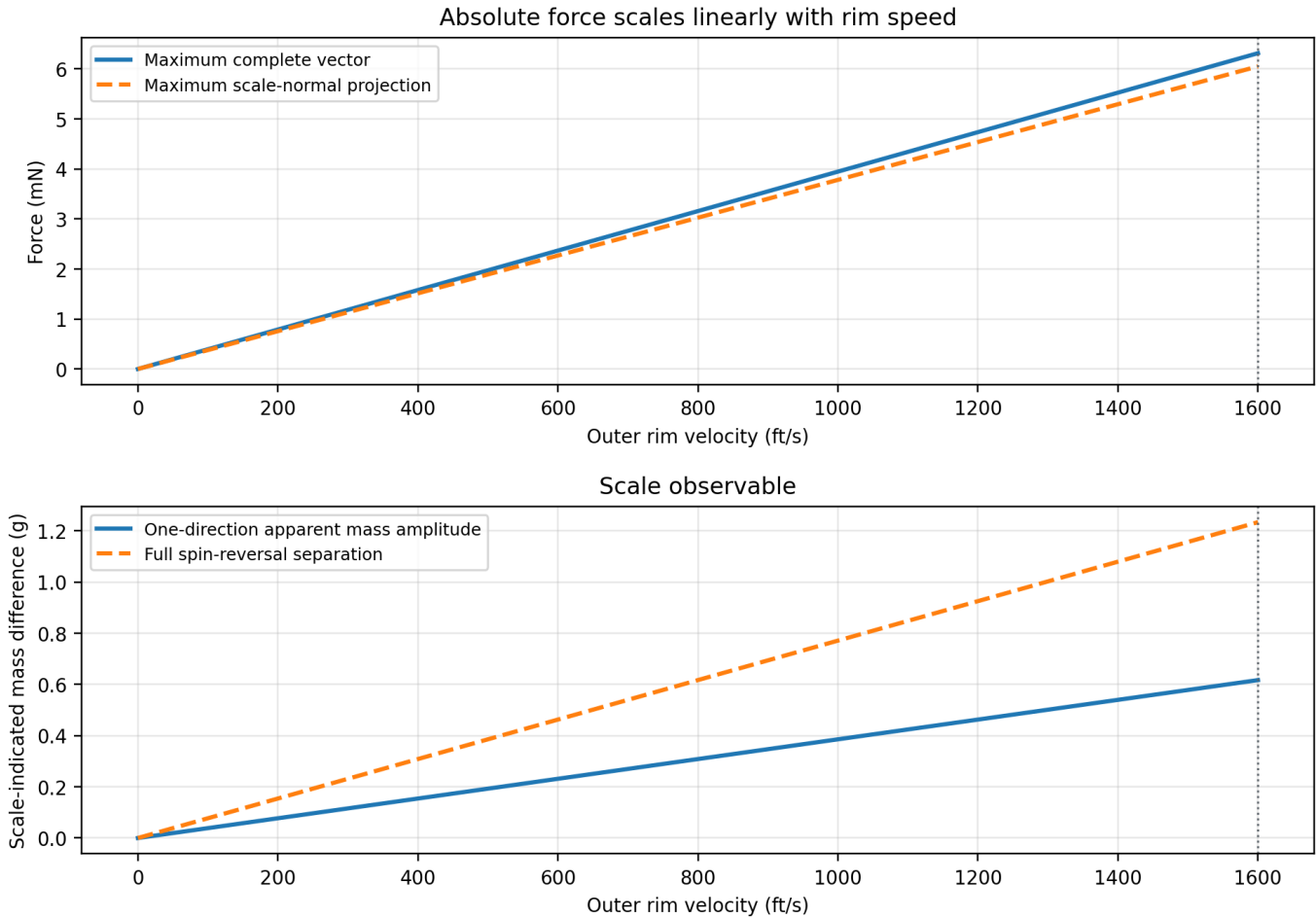


Figure 6. Maximum complete force, maximum fixed-scale projection, one-direction apparent mass, and full reversal separation as functions of outer-rim velocity. The response is linear in rim velocity because $\omega = v_o/r_o$.

At the optimum unrestricted orientation, the computed acceleration and force magnitudes are

Quantity	Nominal	Coupling interval
Maximum acceleration	1.395449549837 m/s ²	[1.381236103342, 1.409662996333] m/s ²
Maximum complete force	6.312865038 mN	[6.248564921, 6.377165155] mN
Acceleration in standard g	0.142296253036 g	nominal
Force / rotor weight	14.229625304%	nominal

Using the source force-direction unit vector, the ICRS vectors are

```
a_K = (0.108104898103, 1.042925312089, 0.920814623338) m/s^2
F_K = (4.890550373222e-04, 4.718082958128e-03, 4.165667216635e-03) N
```

9. Fixed-laboratory scale projection

A weigh scale measures the support force along the local vertical. For local upward unit vector U , the apparent scale force is

$$\begin{aligned} W_{\text{scale}} &= M_{\text{assembly}} g_0 - F_K \cdot U \\ \Delta W &= -F_U \\ F_U &= (\kappa_F m \omega / 2) (U \times n) \cdot g \end{aligned}$$

The optimum passive fixed-laboratory orientation is a horizontal rotor axis aligned true north-south. The local east vector then sweeps the full celestial equatorial plane once per sidereal day. The maximum scale projection is the equatorial magnitude of g divided by the complete magnitude of g .

Quantity	Result
Equatorial response magnitude	891908.761324178427
Complete response magnitude	930825.217270434834
Projection factor	0.958191446445
Maximum scale-normal acceleration	1.337107822600 m/s ²
Maximum scale-normal force	6.048933282 mN
Equivalent apparent mass amplitude	0.616819534 g
Coupling interval apparent mass	[0.610536876, 0.623102191] g

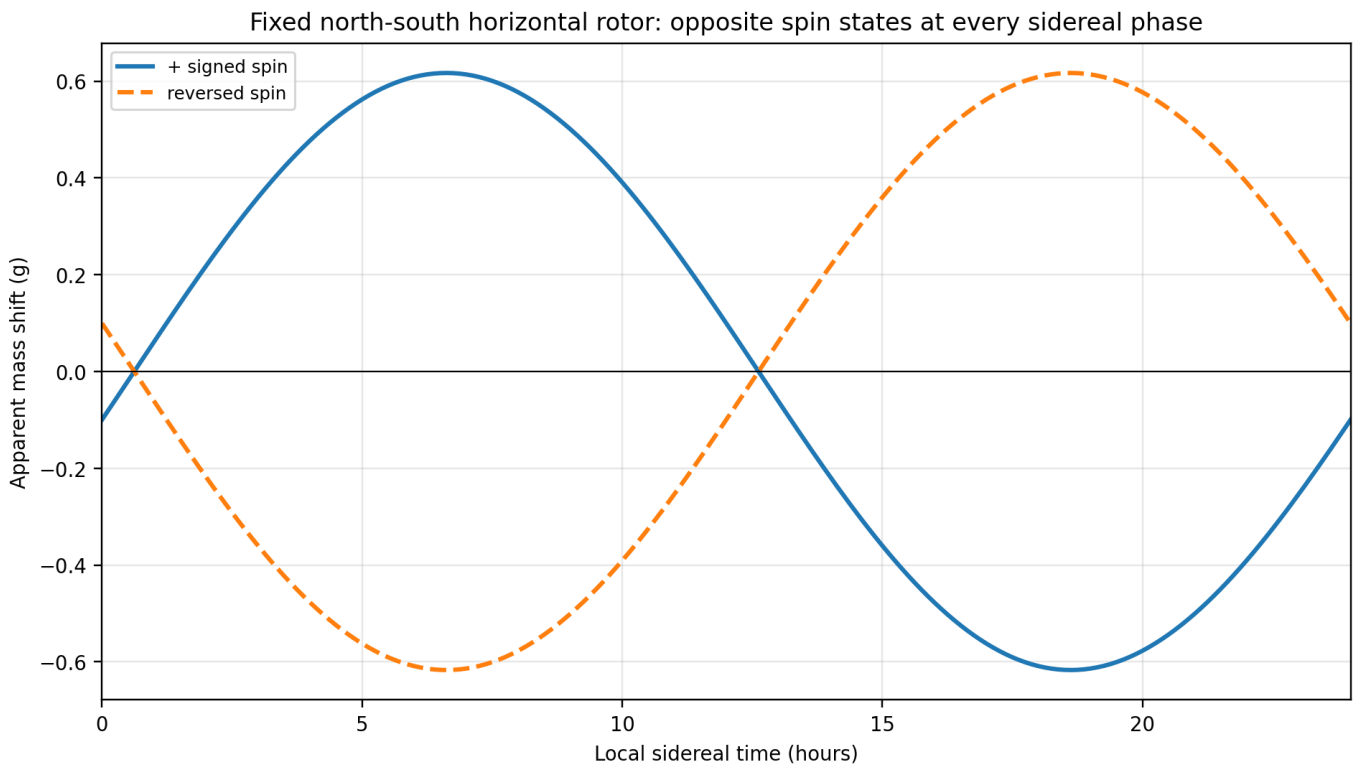


Figure 7. At every sidereal phase, reversing axial spin changes the sign of the complete scale signal. The largest separation occurs at the two phases where the vertical projection reaches its extrema.

10. Force as a percentage of rotor tube weight

Force and mass have different dimensions. The dimensionless percentage requested for the rotor is therefore formed by dividing force by the tube gravitational weight $m g_0$.

$$P_F = F / (m g_0) \times 100\%$$

$$P_U = F_U / (m g_0) \times 100\%$$

$$P_{\text{reversal}} = 2 F_U / (m g_0) \times 100\%$$

Measure	Force	Equivalent mass	Percent of tube weight
Maximum complete vector	6.312865038 mN	0.643733083 g	14.229625304%
Maximum scale-normal state	6.048933282 mN	0.616819534 g	13.634705252%
Full +omega to -omega reversal	12.097866564 mN	1.233639068 g	27.269410504%

The tube gravitational weight is 0.044364239419 N. One optimum scale state is 13.634705% of tube weight, and the full controlled spin-reversal separation is 27.269411% of tube weight.

Because both F_U and $m g_0$ are proportional to mass, the force-to-weight percentage is independent of density and tube length for fixed radius, field orientation, and angular speed. Carbon fiber reduces absolute force relative to an equal-speed steel tube, but it does not reduce the normalized acceleration or force-to-weight ratio.

11. Controlled spin-reversal experiment

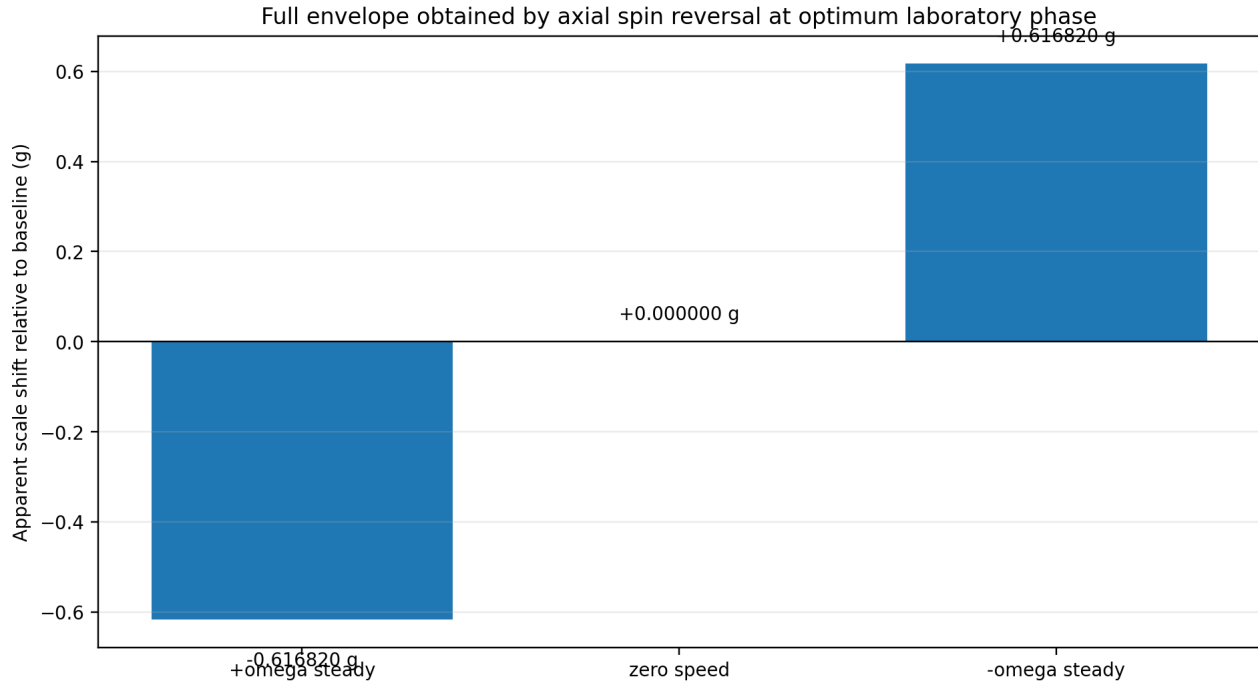


Figure 8. At the optimum laboratory phase, equal positive and negative angular speeds place the scale at opposite ends of the full envelope. The zero-speed state is the midpoint.

At fixed laboratory geometry, reversing the tube direction changes ω to $-\omega$ and therefore reverses the complete force vector. The two steady scale readings are

$$\begin{aligned} W(+) &= M_{\text{assembly}} g_0 - F_U \\ W(-) &= M_{\text{assembly}} g_0 + F_U \\ W(-) - W(+) &= 2 F_U \end{aligned}$$

State	Scale shift relative to nonrotating baseline
+omega steady	-0.616819534 g
Zero speed	+0.000000000 g
-omega steady	+0.616819534 g
Opposite-spin separation	1.233639068 g

This reproduces the two extrema that Earth rotation would otherwise present approximately 12 sidereal hours apart. The complete global maximum is obtained when the laboratory is near the optimum vertical-projection phase. At other times, reversal still produces the full instantaneous separation $2|F_U(t)|$.

The clean observable is the spin-odd channel

$$S_{\text{odd}} = [W(+\omega) - W(-\omega)] / 2 = -F_U$$

Equal-magnitude opposite-spin measurements reject scale offset, assembly weight, gravitational tides over short reversal intervals, and other predominantly spin-even backgrounds. Data used for the comparison should be acquired only after both directions reach the same stable speed magnitude and transient torque, vibration, and thermal responses have settled.

12. Comparison with the stainless reference rotor

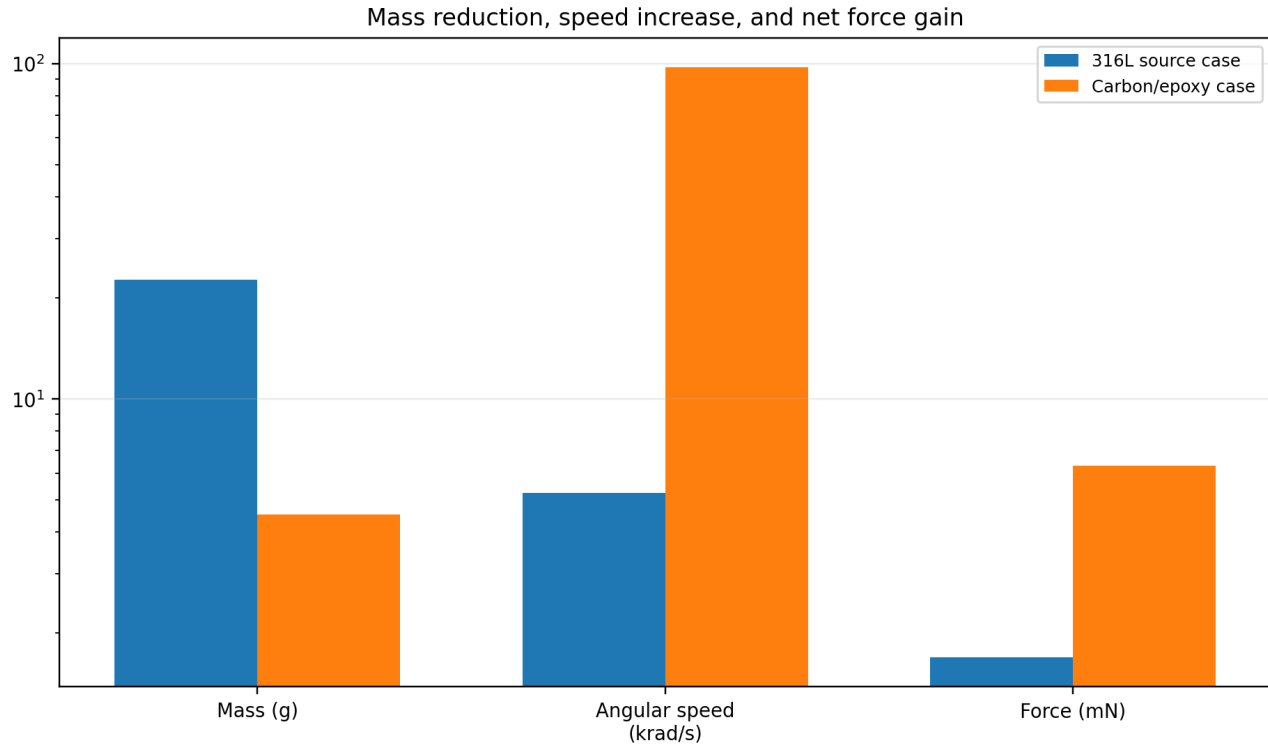


Figure 9. Logarithmic comparison of source stainless mass and speed with the carbon/epoxy operating state. The carbon tube has one-fifth the mass, 18.628 times the angular speed, and 3.7256 times the maximum force.

Quantity	316L reference	Carbon/epoxy experiment	Ratio CF / steel
Density	8000.0 kg/m ³	1600.0 kg/m ³	0.200000000
Tube mass	22.619467106 g	4.523893421 g	0.200000000
Rotation rate	50,000.000 rpm	931,400.192 rpm	18.628003835
Angular speed	5235.987755983 rad/s	97536.000000000 rad/s	18.628003835
Maximum force	1.694455588 mN	6.312865038 mN	3.725600767
Maximum fixed-scale force	1.623612851 mN	6.048933282 mN	3.725600767

Since force is proportional to $m \omega$, the exact force ratio is the product of the mass ratio and angular-speed ratio:

$$F_{CF} / F_{steel} = (m_{CF}/m_{steel})(\omega_{CF}/\omega_{steel})$$

$$= 0.200000000000 \times 18.628003835293 = 3.725600767059$$

13. Scaling laws and material implications

For fixed field orientation and identical tube radius, the principal scaling laws are

```
m proportional to rho L (r_o^2-r_i^2)
omega = v_o/r_o
F_max proportional to rho L (r_o^2-r_i^2) v_o/r_o
a_max proportional to v_o/r_o
E_spin proportional to rho L (r_o^2-r_i^2) v_o^2
F/(m g0) = a/g0
```

These relations separate absolute-force design from normalized response. Increasing density or tube length increases absolute force linearly but leaves free acceleration and force-to-weight percentage unchanged. Increasing rim speed increases K-Field force linearly, while rotational energy increases quadratically. Reducing outer radius at fixed rim speed increases angular speed and predicted acceleration as $1/r_o$ under the present law.

The carbon/epoxy case therefore gains force over the slower steel reference because the 18.628-fold speed increase is larger than the fivefold mass reduction. The energy cost grows more rapidly than force because energy is quadratic in speed.

Design variable	Absolute force	Acceleration / force-to-weight	Spin energy
Density rho	linear	no change	linear
Length L	linear	no change	linear
Signed angular speed omega	linear with sign	linear with sign	quadratic
Outer rim velocity at fixed radius	linear	linear	quadratic
Number of aligned same-spin tubes	linear	assembly-normalized value depends on inactive mass	linear

14. Measurement architecture and controls

The recommended laboratory configuration places the rotor axis horizontally along true north-south and mounts the complete motor, bearings, containment, and rotor on a force scale. The scale should be sampled continuously while rotor speed, motor current, bearing temperature, vibration, and local sidereal time are recorded.

Stage	Action	Measurement purpose
1	Record nonrotating baseline	Establish assembly weight, scale drift, and thermal baseline
2	Accelerate to $+\omega$ and allow settling	Reach constant +931,400.2 rpm state
3	Acquire steady $+\omega$ interval	Measure one polarity of the spin-odd signal
4	Decelerate through zero	Exclude transient interval from force comparison
5	Accelerate to $-\omega$ with equal magnitude	Reverse only the signed angular speed
6	Allow thermal and vibration settling	Match steady operating conditions
7	Acquire steady $-\omega$ interval	Measure opposite force polarity
8	Compute half-difference	Extract F_U from the spin-odd channel
9	Repeat over sidereal phase	Verify predicted daily amplitude and phase

Critical controls

Control	Reason
Equal $ \omega $ in both directions	The signal is linear in signed angular speed
Identical averaging intervals	Prevents unequal drift weighting
Bearing and motor reversal characterization	Separates mechanical direction-dependent reaction
Independent vibration sensor	Rejects scale rectification by vibration
Thermal stabilization	Controls motor and bearing heat drift
Dummy nonrotating mass and powered motor control	Separates electromagnetic and drive-system effects
Vertical-axis null run	The direct scale projection is zero when n is parallel to U
North-south versus east-west orientation run	Tests the directional transformation
Multiple rotor speeds	Tests linear force versus ω
Local sidereal timestamping	Tests the inertial orientation waveform

The speed and stored energy require a purpose-built high-speed rotor system and containment. The experiment should not be attempted with an uncertified tube, improvised bearings, or uncontained rotor.

15. Findings and conclusions

Finding	Result
1	The identical carbon/epoxy tube geometry contains $2.827433388230815 \times 10^{-6} \text{ m}^3$ of composite and has mass 4.523893421 g at 1600 kg/m^3 .
2	A 1600 ft/s outer-rim speed at 5 mm radius requires 97536.000 rad/s and $931,400.2 \text{ rpm}$.
3	The axial moment of inertia is $9.273981513397073 \times 10^{-8} \text{ kg m}^2$, angular momentum is $0.009045470609 \text{ kg m}^2/\text{s}$, and stored rotational energy is 441.129510655 J .
4	The maximum complete K-Field acceleration is $1.395449550 \text{ m/s}^2$ and the maximum complete force is 6.312865038 mN .
5	The optimum fixed-scale projection is 6.048933282 mN , equivalent to 0.616819534 g in one direction.
6	Controlled spin reversal produces a full 12.097866564 mN or 1.233639068 g separation at the optimum laboratory phase.
7	The scale-normal force is 13.634705252% of rotor tube weight, while full reversal spans 27.269410504% of rotor tube weight.
8	The carbon/epoxy experiment produces 3.725600767 times the maximum force of the $50,000 \text{ rpm}$ stainless reference because its 0.2 mass ratio is multiplied by an 18.628003835 angular-speed ratio.
9	The normalized force-to-weight result is independent of material density under the stated law; density changes absolute force and spin energy, not acceleration.
10	Reversal provides the fastest complete-envelope demonstration, while the sidereal sweep supplies the directional and inertial-frame confirmation.

The carbon-fiber tube calculation establishes a direct quantitative target for the fixed-laboratory experiment. Its predicted scale signal is substantially larger than the original stainless 50,000 rpm case, but the required rotational state is correspondingly severe. The K-Field measurement and the mechanical rotor qualification are separate engineering tasks and must both be satisfied.

Appendix A. Complete numerical input ledger

Response matrix

Row	G_phase col 1	G_phase col 2	G_phase col 3
1	2.976410637800	1.243150779500	-2.080360035400
2	1.243150779500	2.592228663500	-2.317883993700
3	-2.080360035400	-2.317883993700	6.773570913300

Scalar and vector inputs

Field	Value
Source schema	K-Field.tubular_rotor.free_space_forces.v1
Source epoch	1784904809
Source title	K-Field Tubular Rotor Free-Space Forces
Carbon material record	Goodfellow Carbon/Epoxy Composite Tube
Carbon density kg/m ³	1600.0
Outer diameter m	0.01
Inner diameter m	0.008
Wall thickness m	0.001
Length m	0.1
Volume m ³	2.82743338823081459e-06
Mass kg	4.52389342116930361e-03
Rim velocity ft/s	1600.0
Rim velocity m/s	487.68
Omega rad/s	97536.0
RPM	931400.1917646662
Inertia kg m ²	9.27398151339707271e-08
Angular momentum kg m ² /s	9.04547060890696815e-03
Spin energy J	4.41129510655175068e+02
Maximum force N	6.31286503808334114e-03
Maximum vertical force N	6.04893328205544610e-03
One-direction apparent mass g	0.616819533893373
Full reversal apparent mass g	1.233639067786746
Vertical force percent weight	13.63470525205127
Full reversal percent weight	27.26941050410254

Appendix B. All 27 Cell3 address nodes

Integer address	ICRS x native	ICRS y native	ICRS z native	Class	Index sum
(-1, -1, -1)	-0.010974902499961	0.007385983127423	0.039661911011939	corner	-3
(-1, -1, 0)	-0.011290092242635	-0.004742877881141	0.018199733726392	edge center	-2
(-1, -1, 1)	-0.011605281985309	-0.016871738889705	-0.003262443559155	corner	-1
(-1, 0, -1)	-0.009875529515800	0.013837068256769	0.022628660243911	edge center	-2
(-1, 0, 0)	-0.010190719258473	0.001708207248205	0.001166482958364	face center	-1
(-1, 0, 1)	-0.010505909001147	-0.010420653760359	-0.020295694327182	edge center	0
(-1, 1, -1)	-0.008776156531638	0.020288153386115	0.005595409475884	corner	-1
(-1, 1, 0)	-0.009091346274312	0.008159292377552	-0.015866767809663	edge center	0
(-1, 1, 1)	-0.009406536016985	-0.003969568631012	-0.037328945095210	corner	1
(0, -1, -1)	-0.000784183241488	0.005677757879218	0.038495428053575	edge center	-2
(0, -1, 0)	-0.001099372984162	-0.006451085129346	0.017033250768028	face center	-1
(0, -1, 1)	-0.001414562726835	-0.018579946137910	-0.004428926517519	edge center	0
(0, 0, -1)	0.000315189742674	0.012128861008564	0.021462177285547	face center	-1
(0, 0, 0)	0.000000000000000	0.000000000000000	0.000000000000000	body center	0
(0, 0, 1)	-0.000315189742674	-0.012128861008564	-0.021462177285547	face center	1
(0, 1, -1)	0.001414562726835	0.018579946137910	0.004428926517519	edge center	0
(0, 1, 0)	0.001099372984162	0.006451085129346	-0.017033250768028	face center	1
(0, 1, 1)	0.000784183241488	-0.005677757879218	-0.038495428053575	edge center	2
(1, -1, -1)	0.009406536016985	0.003969568631012	0.037328945095210	corner	-1
(1, -1, 0)	0.009091346274312	-0.008159292377552	0.015866767809663	edge center	0
(1, -1, 1)	0.008776156531638	-0.020288153386115	-0.005595409475884	corner	1
(1, 0, -1)	0.010505909001147	0.010420653760359	0.020295694327182	edge center	0
(1, 0, 0)	0.010190719258473	-0.001708207248205	-0.001166482958364	face center	1
(1, 0, 1)	0.009875529515800	-0.013837068256769	-0.022628660243911	edge center	2
(1, 1, -1)	0.011605281985309	0.016871738889705	0.003262443559155	corner	1
(1, 1, 0)	0.011290092242635	0.004742877881141	-0.018199733726392	edge center	2
(1, 1, 1)	0.010974902499961	-0.007385983127423	-0.039661911011939	corner	3

Authoritative Cell3 base cell: six faces, 12 edges, all 27 nodes

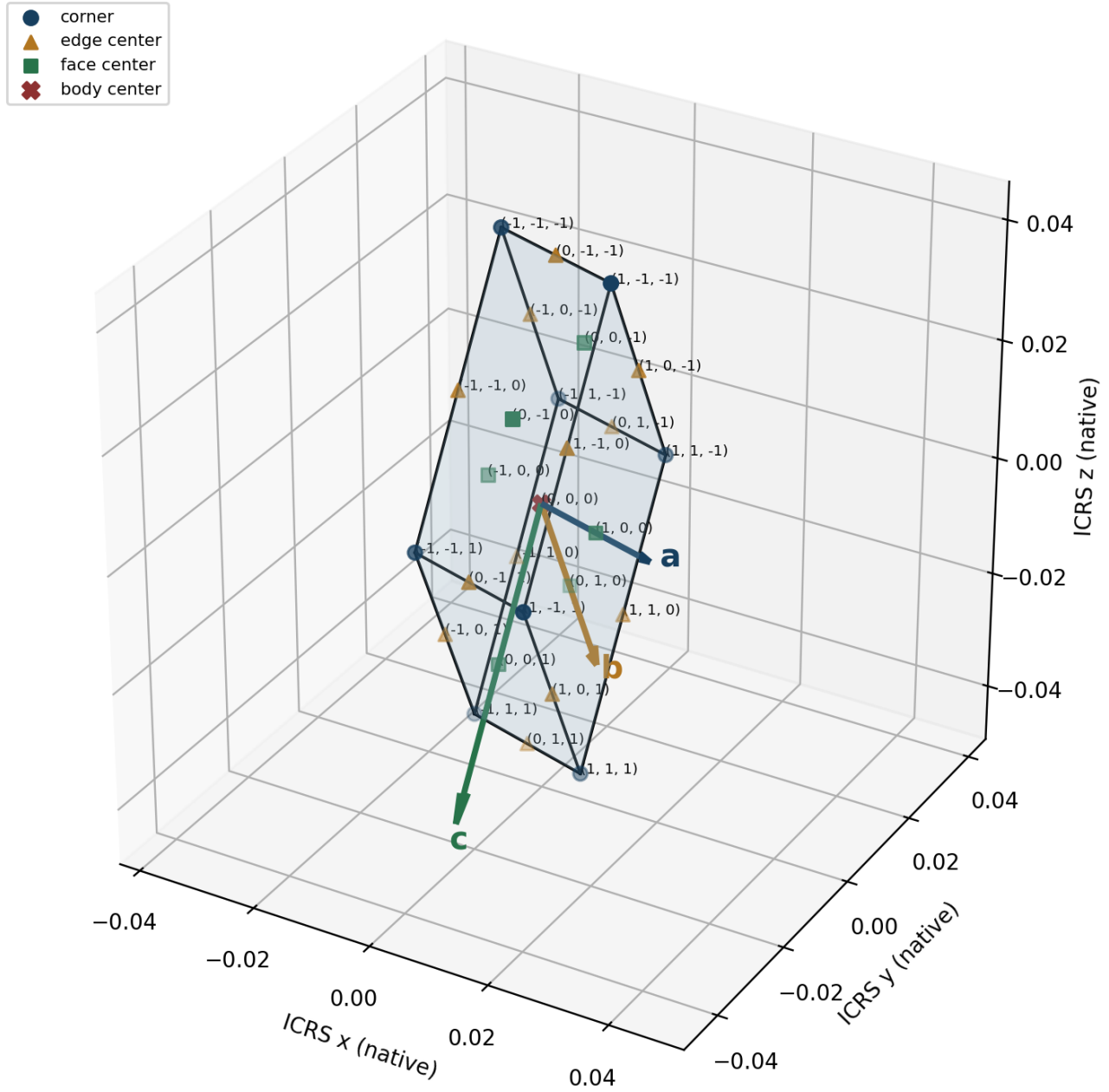


Figure B1. All node coordinates and basis vectors are rendered directly from the source ledger.

Appendix C. Velocity and force sweep

The table lists 100 ft/s increments from rest to the requested 1600 ft/s operating point. The machine-readable companion contains 10 ft/s increments.

Rim ft/s	RPM	Force mN	Scale force mN	One-state g	Scale force / weight %	Energy J
0	0.0	0.000000	0.000000	0.000000	0.000000	0.000000
100	58,212.5	0.394554	0.378058	0.038551	0.852169	1.723162
200	116,425.0	0.789108	0.756117	0.077102	1.704338	6.892649
300	174,637.5	1.183662	1.134175	0.115654	2.556507	15.508459
400	232,850.0	1.578216	1.512233	0.154205	3.408676	27.570594
500	291,062.6	1.972770	1.890292	0.192756	4.260845	43.079054
600	349,275.1	2.367324	2.268350	0.231307	5.113014	62.033837
700	407,487.6	2.761878	2.646408	0.269859	5.965184	84.434945
800	465,700.1	3.156433	3.024467	0.308410	6.817353	110.282378
900	523,912.6	3.550987	3.402525	0.346961	7.669522	139.576134
1000	582,125.1	3.945541	3.780583	0.385512	8.521691	172.316215
1100	640,337.6	4.340095	4.158642	0.424063	9.373860	208.502620
1200	698,550.1	4.734649	4.536700	0.462615	10.226029	248.135350
1300	756,762.7	5.129203	4.914758	0.501166	11.078198	291.214404
1400	814,975.2	5.523757	5.292817	0.539717	11.930367	337.739782
1500	873,187.7	5.918311	5.670875	0.578268	12.782536	387.711484
1600	931,400.2	6.312865	6.048933	0.616820	13.634705	441.129511

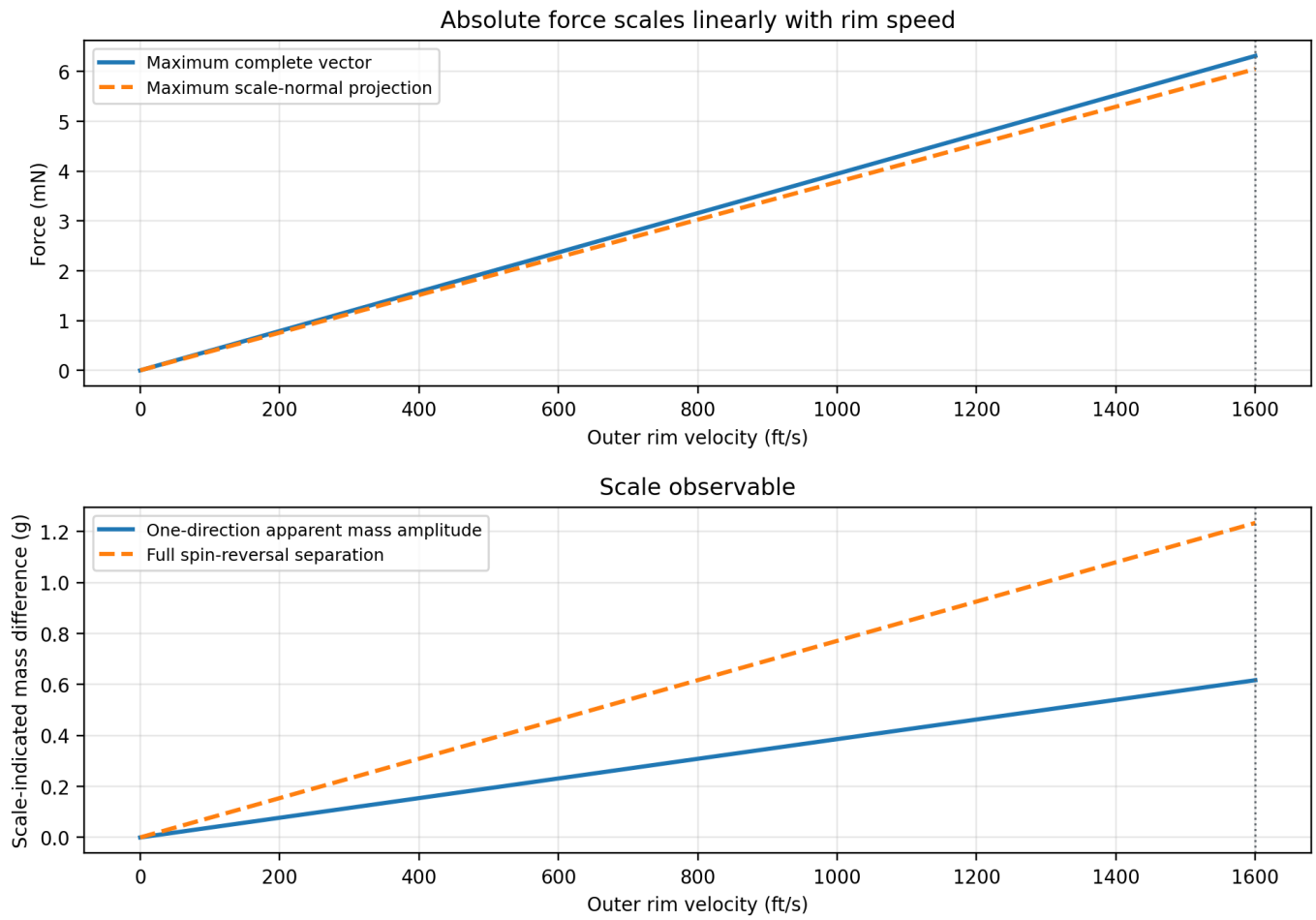


Figure C1. Continuous force and scale-observable curves used to interpolate the tabulated sweep.

Appendix D. Compact reconstruction equations

Step	Equation
1. Tube area	$A = \pi(r_o^2 - r_i^2)$
2. Tube volume	$V = A L$
3. Tube mass	$m = \rho V$
4. Rim conversion	$v_o = 1600 \text{ ft/s} \times 0.3048$
5. Angular speed	$\omega = v_o / r_o$
6. Rotation rate	$\text{RPM} = \omega 60 / (2 \pi)$
7. Axial inertia	$I = (m/2)(r_o^2 + r_i^2)$
8. Angular momentum	$H = I \omega$
9. Spin energy	$E = (1/2) I \omega^2$
10. Response vector	$g = G_{\text{phase}} V_E$
11. Tube force	$F_K = (\kappa_F m \omega / 2)(n \times g)$
12. Maximum acceleration	$a_{\text{max}} = (\kappa_F \omega / 2) g $
13. Fixed-scale projection	$F_{U,\text{max}} = F_{\text{max}} g_{xy} / g $
14. Apparent mass	$\Delta m = F_U / g_0$
15. Force percentage	$P_U = F_U / (m g_0) \times 100\%$
16. Reversal separation	$\Delta F_{\text{rev}} = 2F_U; \Delta m_{\text{rev}} = 2\Delta m$

Reconstruction procedure

Read the tube dimensions and commercial density. Compute annular volume and mass. Convert 1600 ft/s to SI velocity, then compute ω and RPM from the 5 mm radius. Compute inertia, angular momentum, and energy. Read G_{phase} , V_E , and κ_F from the source ledger. Form $g = G_{\text{phase}} V_E$. Evaluate maximum acceleration and force, apply the fixed-scale equatorial projection factor, convert force to apparent mass, and normalize by $m g_0$. For spin reversal, change only the sign of ω .

Appendix E. Archival and source record

Field	Record
Document title	K-Field Carbon-Fiber Tube Laboratory Experiment
Principal Investigator	P. Dwayne Esterline
Education	BS, Huntington University (1999); MBA, Indiana Wesleyan University (2008)
Organization	Krytronx
Classification	Proprietary Confidential Information and Trade Secret of Krytronx, LLC. All Rights Reserved. Do Not Distribute.
Generation UTC	2026-07-24T17:55:45Z
Generation epoch	1784915745
Output PDF	K-Field_carbon_fiber_tube_lab_experiment_1784915745.pdf
Machine-readable companion	K-Field_carbon_fiber_tube_lab_experiment_1784915745.json
Primary source JSON	K-Field_tubular_rotor_free_space_forces_1784904809(1).json
Material density source	Goodfellow Carbon/Epoxy Composite Tube; MatWeb; density 1.60 g/cc
Material source URL	https://www.matweb.com/search/DataSheet.aspx?MatGUID=51313a82bc67483abf4f52c0e2934c88
Historical format references	K-Field Tubular Rotor Free-Space Forces; K-Field Apparent-Weight Modulation Spin-Reversal Experiment; K-Field Dual-Axis Kinetic Mass CMB Traversal Geometry
Coordinate frame	ICRS/J2000 Cartesian; local east-north-up scale projection
Mechanical units	SI; rim speed also in ft/s; equivalent scale mass in grams
Cell3 units	native source units
Report generation	Python ReportLab; built-in core fonts only
Figure generation	Python numerical computation and raster rendering from exact report ledger
Page size	US Letter
Margins	0.68 inch
Table widths	dynamically normalized to available page width
Cell wrapping	every table text cell is a ReportLab Paragraph
Long-table headers	repeatRows=1

P. Dwayne Esterline
Krytronx

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